

An Efficient Approach for Cell Tower Performance Measurement Using Autonomous Aerial Vehicle

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Abstract—In an era where automation and cost efficiency are held on a pedestal, unmanned aerial vehicles (UAVs) are becoming a practical tool for high-resolution measurement of cellular network performance. They enable flexible, low-cost data collection across complex geographic regions. This work proposes spatial grid aggregation, correlation-based master grid selection, and ACO-based path and energy optimization.

I. INTRODUCTION

UAVs enable flexible, low-cost data collection for cellular network measurements. Dense UAV datasets create redundancy, spatial noise, and high computational cost. To address this, bin-based grid quantization, correlation analysis, predictive modeling, and ACO-based route optimization are applied.

II. PROPOSED WORK

A. Grid Formation

The target area is divided into fixed grids using latitude and longitude binning. Each grid is assigned a unique ID and grids with insufficient data are removed.

B. Master-Slave Grid Selection

A correlation matrix is generated between grids. A threshold-based cumulative correlation method selects master grids and assigns correlated grids as slaves to reduce redundant measurements.

C. Path Planning and Distance Optimization

The centroid of each grid is computed as:

$$C = \left(\frac{lat_{min} + lat_{max}}{2}, \frac{lon_{min} + lon_{max}}{2} \right)$$

Latitude and longitude are converted into planar coordinates:

$$x_i = (lon_i - lon_0) \times 111.32 \cos(lat_0)$$

$$y_i = (lat_i - lat_0) \times 110.574$$

Distance between nodes:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$

Algorithm 1 Path Planning

- 1: Load unique grid centroids $V \leftarrow \{(lat_i, lon_i)\}$
- 2: Convert latitude-longitude to planar coordinates using local scaling
- 3: Construct complete graph $G = (V, E)$
- 4: Compute edge weights:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$

- 5: Initialize parameters: $m = |V|$, $T = 1600$, $\alpha = 1$, $\beta = 2$, $\rho = 0.5$
- 6: Initialize pheromone: $\tau_{ij} = 1$
- 7: Apply Ant Colony Optimization:

$$P^* \leftarrow ACO(G)$$

- 8: Compute total distance:

$$D_{total} = \sum_{(i,j) \in P^*} d_{ij}$$

D. Energy Optimization

UAV energy consumption includes distance-based energy and turning energy. Each node is represented as (x_i, y_i, z_i) where z_i is the altitude.

Distance-based energy:

$$E_{ij}^{dist} = P_{cruise} \frac{d_{ij}}{V} + C_{corr}$$

Deviation angle for transition $i \rightarrow j \rightarrow k$:

$$\vec{v}_{ij} = (x_j - x_i, y_j - y_i, z_j - z_i)$$

$$\vec{v}_{jk} = (x_k - x_j, y_k - y_j, z_k - z_j)$$

$$\theta_{ijk} = \cos^{-1} \left(\frac{\vec{v}_{ij} \cdot \vec{v}_{jk}}{\|\vec{v}_{ij}\| \|\vec{v}_{jk}\|} \right)$$

Turning energy:

$$E_{ijk}^{turn} = C_{turn} \frac{\theta_{ijk}}{\theta_{ref}}$$

Total transition energy:

$$E_{ijk} = E_{ij}^{dist} + E_{ijk}^{turn}$$

Algorithm 2 Energy Optimization

- 1: Perform Steps 1–4 of Algorithm 1
- 2: Initialize UAV parameters: $V = 2$ m/s, $P_{cruise} = 308.709$ W, $C_{turn} = 225$, $\theta_{ref} = 2.1$ rad, $C_{corr} = -0.852$
- 3: Compute distance-based energy E_{ij}^{dist}
- 4: Compute deviation angle θ_{ijk}
- 5: Compute turning energy E_{ijk}^{turn}
- 6: Compute total transition energy E_{ijk}
- 7: Initialize ACO parameters: $m = |V|$, $T = 1600$, $\alpha = 1$, $\beta = 2$, $\rho = 0.5$, $Q = 1$
- 8: Initialize pheromone: $\tau_{ij} = 1$
- 9: Construct energy-weighted graph where $w(i, j, k) = E_{ijk}$
- 10: Solve energy-constrained TSP:

$$P^* \leftarrow ACO(G)$$

- 11: Compute total distance:

$$D_{total} = \sum_{(i,j) \in P^*} d_{ij}$$

- 12: Compute total energy:

$$E_{total} = \sum_{(i,j,k) \in P^*} E_{ijk}$$

E. Final Energy Calculation

For a UAV traversal path

$$P^* = \{n_0, n_1, \dots, n_m\}$$

$$E_{total} = \sum_{i=0}^{m-1} \left(P_{cruise} \frac{d_{i,i+1}}{V} + C_{corr} \right) + \sum_{i=1}^{m-1} \left(C_{turn} \frac{\theta_i}{\theta_{ref}} \right)$$

where $d_{i,i+1}$ is the distance between consecutive nodes (meters) and θ_i is the deviation angle (radians).

Parameters used:

$$V = 2 \text{ m/s}, \quad P_{cruise} = 308.709 \text{ W}, \quad C_{turn} = 225, \quad \theta_{ref} = 2.1, \quad C_{corr} = -0.852$$